

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$\frac{1}{2}y=4$$

$$x-\frac{1}{2}y=2$$

The system of equations above has solution (x, y). What is the value of x ?

Answer

- A. 3
- B. $\frac{7}{2}$
- C. 4
- D. 6

Correct Answer: D

Rationale

Choice D is correct. Adding the corresponding sides of the two equations eliminates y and yields $x = 6$, as shown.

$$\frac{1}{2}y=4$$

$$x-\frac{1}{2}y=2$$

$$\hline x+0=6$$

If (x, y) is a solution to the system, then (x, y) satisfies both equations in the system and any equation derived from them. Therefore, $x = 6$.

Choices A, B, and C are incorrect and may be the result of errors when solving the system.